Database Design Reasoning – F&O Trading Data

1. Design Choices

The database schema was designed to store and analyze high-volume Futures and Options trading data from multiple exchanges including NSE, BSE, and MCX.

The dataset contains over 2.5 million records with attributes such as instrument type, symbol, expiry date, strike price, option type, OHLC prices, open interest, and timestamp.

To support efficient analytics and maintain data integrity, the schema was designed following **Third Normal Form (3NF)**.

Normalization ensures that:

- Data redundancy is minimized
- Each column stores atomic values
- Data consistency is maintained across records

The central table in the schema is the **Trades table**, which stores time-series trading information including prices, volumes, and open interest.

The schema also includes attributes such as:

- Exchange
- Instrument
- Symbol
- Expiry date
- Strike price

This structure supports cross-exchange analysis and option chain analytics.

2. Table Structure

The primary table implemented is:

Trades Table

Key columns include:

- instrument – Type of financial instrument (FUTIDX, OPTIDX etc.)
- exchange – Exchange name such as NSE, BSE, or MCX
- symbol – Trading symbol
- expiry_dt – Expiry date of the contract
- strike_pr – Strike price for options

- option_typ – Call or Put option
- open, high, low, close – Daily OHLC prices
- settle_pr – Settlement price
- contracts – Number of contracts traded
- open_int – Open interest
- chg_in_oi – Change in open interest
- timestamp – Trade date

Numeric data types are used for price values and volumes to maintain precision in financial calculations.

3. Query Optimization

Since the dataset contains millions of rows, several optimization strategies were implemented.

Indexing

Indexes were created on frequently queried columns:

- symbol
- timestamp

These indexes improve performance for:

- filtering by symbol
- time-series analysis
- rolling window calculations

Window Functions

Advanced SQL window functions were used to perform calculations such as:

- 7-day rolling volatility
- ranking symbols by trading volume

Window functions allow efficient time-series analytics without complex joins.

Query Performance Analysis

The EXPLAIN ANALYZE command was used to evaluate query execution plans and measure performance improvements after indexing.

4. Scalability

The database design is scalable for handling **10 million or more trading records**.

Key scalability strategies include:

- Time-based indexing
- Partitioning strategies based on expiry date or exchange
- Efficient aggregation queries
- Reduced data redundancy through normalization

These optimizations ensure that the system can support high-frequency trading analytics and large financial datasets.